AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	CyrusOne Austin Data Center II, TX -- station KAUS, America/Chicago
Committed pair	OSM 455171103 -> 379204766 CyrusOne Austin Data Center III / CyrusOne Austin Data Center II
Facade gap	149.9 m between the two halls
Weather record	43,746 real hourly records from KAUS
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.4627 C at 285 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-10-29 (knife_edge)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 5 of 24 hours with 1 mode change. The reactive incumbent took 8 hours -- 3 more -- but declared 0 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 8 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	5 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	8 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 8 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	15.560	18.0	15.944	1.048
01	mechanical	yes	switch budget	15.275	18.0	14.587	1.201
02	mechanical	yes	switch budget	15.275	18.0	14.823	1.036
03	mechanical	yes	switch budget	14.737	18.0	13.713	0.894
04	mechanical	yes	switch budget	14.445	18.0	14.311	0.754

05	mechanical	yes	switch budget	14.720	18.0	14.747	0.668
06	mechanical	yes	switch budget	14.557	18.0	14.747	0.846
07	mechanical	yes	switch budget	15.280	18.0	14.600	0.883
08	mechanical	no	dry-bulb	18.055	18.0	15.708	1.681
09	mechanical	no	dry-bulb	20.912	18.0	17.276	2.623
10	mechanical	no	dry-bulb	22.221	18.0	18.386	2.978
11	mechanical	no	dry-bulb	22.514	18.0	20.601	1.910
12	mechanical	no	dry-bulb	22.780	18.0	21.681	1.173
13	mechanical	no	dry-bulb	23.476	18.0	22.231	1.058
14	mechanical	no	dry-bulb	24.579	18.0	23.330	0.919
15	mechanical	no	dry-bulb	24.170	18.0	23.378	0.852
16	mechanical	no	dry-bulb	23.889	18.0	23.463	0.858
17	mechanical	no	dry-bulb	23.793	18.0	22.518	1.117
18	mechanical	no	dry-bulb	21.386	18.0	19.513	1.379
19	FREE	yes	-	17.440	18.0	12.495	1.530
20	FREE	yes	-	15.751	18.0	10.196	1.732
21	FREE	yes	-	14.995	18.0	10.620	1.747
22	FREE	yes	-	10.702	18.0	8.922	1.565
23	FREE	yes	-	10.268	18.0	10.117	1.210

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.560 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.275 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.275 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.737 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.445 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.720 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.557 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.280 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.055 C against a 18.0 C limit, so it fails by 0.055 C. It would take a limit 0.055 C higher, or a bound 0.055 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.912 C against a 18.0 C limit, so it fails by 2.912 C. It would take a limit 2.912 C higher, or a bound 2.912 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.221 C against a 18.0 C limit, so it fails by 4.221 C. It would take a limit 4.221 C higher, or a bound 4.221 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.514 C against a 18.0 C limit, so it fails by 4.514 C. It would take a limit 4.514 C higher, or a bound 4.514 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.780 C against a 18.0 C limit, so it fails by 4.780 C. It would take a limit 4.780 C higher, or a bound 4.780 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.476 C against a 18.0 C limit, so it fails by 5.476 C. It would take a limit 5.476 C higher, or a bound 5.476 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.579 C against a 18.0 C limit, so it fails by 6.579 C. It would take a limit 6.579 C higher, or a bound 6.579 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.170 C against a 18.0 C limit, so it fails by 6.170 C. It would take a limit 6.170 C higher, or a bound 6.170 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.889 C against a 18.0 C limit, so it fails by 5.889 C. It would take a limit 5.889 C higher, or a bound 5.889 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.793 C against a 18.0 C limit, so it fails by 5.793 C. It would take a limit 5.793 C higher, or a bound 5.793 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.386 C against a 18.0 C limit, so it fails by 3.386 C. It would take a limit 3.386 C higher, or a bound 3.386 C tighter, to change this hour.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.440 C, which is 0.560 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.529 C, and the margin is measured, not chosen: 1.529 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.751 C, which is 2.249 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.732 C, and the margin is measured, not chosen: 1.732 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.995 C, which is 3.005 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.747 C, and the margin is measured, not chosen: 1.747 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.702 C, which is 7.298 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.565 C, and the margin is measured, not chosen: 1.565 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.268 C, which is 7.732 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.210 C, and the margin is measured, not chosen: 1.210 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

Generated by AGENTIC-ARBITER/src/report.py. Verified by being read back: the file was reopened after writing and every hour of the schedule, the headline counts and this site's own name were confirmed present.